

A novel STAP algorithm via volume cross-correlation function on the Grassmann manifold

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ABSTRACT

The performance of space-time adaptive processing (STAP) is often degraded by factors such as limited sample size and moving targets. Traditional clutter covariance matrix (CCM) estimation relies on Euclidean metrics, which fail to capture the intrinsic geometric and structural properties of the covariance matrix, thus limiting the utilization of structural information in the data. To address these issues, the proposed algorithm begins by constructing Toeplitz Hermitian positive definite (THPD) matrices from the training samples. The Brauer disc (BD) theorem is then employed to filter out THPD matrices containing target signals, retaining only clutter-related matrices. These clutter matrices undergo eigendecomposition to construct the Grassmann manifold, enabling CCM estimation through the volume cross-correlation function (VCF) and gradient descent method. Finally, the filter weight vector is computed for filtering. By fully leveraging the structural information in radar data, this approach significantly enhances both accuracy and robustness of clutter suppression. Experimental results on simulated and measured data demonstrate superior performance of the proposed algorithm in heterogeneous environments.

1. Introduction

Clutter significantly affects the target detection and identification capabilities of radar systems [1–3]. Therefore, effectively suppressing clutter is crucial for improving the accuracy and reliability of radar systems [4,5]. Space-time adaptive processing (STAP) leverages the spatial and temporal dimensions of radar echoes, allowing for adaptive adjustment of filter parameters based on different environments and mission requirements to suppress clutter [6–10]. Traditional STAP methods often perform poorly in non-homogeneous clutter environments because they typically use a sample covariance matrix (SCM) as a substitute for the clutter covariance matrix (CCM). When the sample size is insufficient, the estimation of the SCM becomes inaccurate, affecting STAP performance [11–13]. In the presence of range-spread targets, which occupy multiple range cells, training samples may contain multiple target signals, leading to inaccurate estimation of the CCM and target self-nulling, which reduces suppression effectiveness [14–16]. Therefore, achieving more accurate CCM estimation in the presence of limited samples and target interference is a key challenge.

Current methods include using the generalized inner product (GIP) technique [17] to filter out training samples contaminated by target

signals, and employing random matrix theory (RMT) with the spiked covariance model to estimate the matrix by determining the eigenvalues of the SCM [18]. The knowledge-aided STAP method utilizes prior knowledge to improve non-homogeneous clutter suppression performance. This method guides the selection of training strategies and samples or constructs the CCM, which is then fused with the estimated CCM to form the final covariance matrix [19–22]. Reference [23] introduces a knowledge-aided Bayesian framework in which covariance matrices are considered as random variables, with partial information about the covariance matrix of the training samples available. In this context, the maximum a posteriori (MAP) estimate for the primary data covariance matrix is derived. Wen et al. combining waveform coding design techniques with the FDA multiple-input multiple-output (FDA-MIMO) radar system, proposed a novel STAP clutter suppression approach, achieving satisfactory results [24–27]. Furthermore, [28] introduces a robust STAP filter based on the volume cross-correlation function (VCF), which improves clutter suppression under unknown clutter distributions. This filter does not require prior information about the clutter distribution, demonstrating robustness in cluttered environments.

Structured matrices are widely used in many fields such as signal and information processing [29–32]. The Newton maximum likelihood

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estimator (NML) proposed in the literature [30] significantly improves the estimation efficiency of the Toeplitz covariance matrix by combining the Newton method and fast Fourier transform acceleration, reduces the iteration complexity, and solves the computational bottleneck of traditional methods. Under low signal-to-noise ratio and few samples, the estimation error of NML is close to the structured Cramer-Rao bound, and combined with the Levinson-Durbin algorithm to ensure positive definiteness. The regularized Burg algorithm combines model-driven and regularization techniques to efficiently estimate the Toeplitz Hermitian positive definite (THPD) covariance matrix [33]. It retains the Toeplitz structure through an autoregressive model and uses regularization to ensure the matrix positive definiteness, solving the problem that traditional methods are prone to produce singular or non-positive definite matrices in high-dimensional small samples. The algorithm is efficient and stable in recursive calculation and is suitable for real-time processing; the regularization term suppresses noise interference and overfitting, and improves the estimation robustness by balancing the bias and variance. By transforming radar data into THPD matrices, smoothing estimations can be achieved to reduce variance and improve stability, which helps enhance detection and estimation performance. However, these methods have limitations [34]: traditional CCM estimation is based on Euclidean metrics, which cannot always maintain positive definiteness under addition and scalar multiplication, making them unsuitable for processing THPD matrices. Moreover, these methods overlook the intrinsic geometric properties and structural features of the covariance matrix, failing to fully exploit the structural information of the data.

Therefore, to address the shortcomings of existing technologies and achieve better clutter suppression, we propose a novel algorithm based on Brauer disc (BD) theorem [35,36] and VCF on the Grassmann manifold (GVCF), referred to as BGVCF. The main contributions are as follows. First, sliding windows are used to determine the test cells, guard cells, and training samples, and the THPD matrix of the training samples is calculated. Next, based on the BD theorem, clutter THPD matrices without target contamination are selected. The matrices are then eigen-decomposed to form the Grassmann manifold [37,38], and the VCF, along with gradient descent, is applied to estimate the CCM. Finally, a space-time steering vector is constructed, and the filter weight vector is calculated. Compared with existing methods, this approach can more accurately and effectively distinguish target signals from clutter, ensuring that covariance matrix estimation is not affected by target interference. Moreover, the method fully utilizes the intrinsic geometric properties and structural features of the covariance matrix, does not rely on specific clutter distribution assumptions, and achieves robust covariance matrix estimation by directly measuring the distance between the sampled signal subspace and the target subspace, leading to effective clutter suppression and significantly improving suppression accuracy and stability.

The remainder of this paper is organized as follows. Section 2 introduces the STAP signal model and related theory. Section 3 provides a detailed implementation of the BGVCF-STAP algorithm. Section 4 reports numerical experiments using both simulated and measured data, demonstrating the effectiveness of the proposed method in clutter suppression. Finally, Section 5 offers concluding remarks and discussions.

2. Signal model and related theory

2.1. Signal model

In this section, we examine an airborne pulse-Doppler radar system equipped with a uniform linear array (ULA) consisting of M elements, each spaced equally by a distance d . During a coherent processing interval (CPI), each element receives N pulses at a constant pulse repetition frequency (PRF). Target detection is framed as a binary hypothesis testing problem [39]. Under hypothesis H_1 , the received signal

includes both the target echo and clutter noise, while under hypothesis H_0 , only clutter and noise are present. The space-time snapshot data $\mathbf{x}(l) \in \mathbb{C}^{M \times 1}$ collected from the l th range cell is represented as

$$\begin{cases} H_0 : \mathbf{x}(l) = \mathbf{c}(l) + \mathbf{n}(l) \\ H_1 : \mathbf{x}(l) = \mathbf{s}(l) + \mathbf{c}(l) + \mathbf{n}(l) \end{cases}, \quad (1)$$

where $\mathbf{s}(l)$ is the target signal, $\mathbf{c}(l)$ is the clutter signal, $\mathbf{n}(l)$ is complex Gaussian noise with a variance of δ^2 . Meanwhile, the clutter signal $\mathbf{c}(l)$ can be expressed as [8]

$$\mathbf{c}(l) = \sum_{i=1}^{N_c} a_{c,i} \mathbf{v}(f_{d,c,i}, f_{s,c,i}), \quad (2)$$

where N_c are the numbers of clutter patches within each range blur, the coefficient $a_{c,i}$ represents the complex scattering amplitude of the clutter patch, $f_{d,c,i}$ and $f_{s,c,i}$ are the corresponding normalized Doppler and spatial frequencies. $\mathbf{v}(f_{d,c,i}, f_{s,c,i}) = \mathbf{v}_d(f_{d,c,i}) \otimes \mathbf{v}_s(f_{s,c,i})$ is the space-time oriented vector of the i th clutter scattering point, $\otimes$ representing the Kronecker product, $\mathbf{v}_d(f_{d,c,i})$ denotes the time-steering vector and $\mathbf{v}_s(f_{s,c,i})$ denotes the space-steering vector, which are defined as

$$\begin{aligned} \mathbf{v}_d(f_{d,c,i}) &= [1, \exp(j2\pi f_{d,c,i}), \dots, \exp(j2\pi(M-1)f_{d,c,i})]^T, \\ \mathbf{v}_s(f_{s,c,i}) &= [1, \exp(j2\pi f_{s,c,i}), \dots, \exp(j2\pi(N-1)f_{s,c,i})]^T. \end{aligned} \quad (3)$$

The snapshot of target $\mathbf{s}(l)$ can be denoted as

$$\mathbf{s}(l) = a_t \mathbf{v}(f_{d,t}, f_{s,t}), \quad (4)$$

where a_t is the reflection coefficient of the target, $\mathbf{v}(f_{d,t}, f_{s,t}) = \mathbf{v}_d(f_{d,t}) \otimes \mathbf{v}_s(f_{s,t})$ is the space-time steering vector of the target, its expression is the same as (3).

The practical approximation of the CCM can be achieved by averaging the autocorrelation of non-target range cell data, as shown below

$$\hat{\mathbf{R}} = \frac{1}{L} \sum_{l=1}^L \mathbf{x}(l) \mathbf{x}^H(l). \quad (5)$$

where L is the number of training samples.

The weighted processing is carried out by taking the inner product of the space-time snapshot data from the current range cell with the corresponding weight vector, which is given by [34]

$$y_l = \mathbf{w}^H \mathbf{x}(l), \quad (6)$$

The normalized mean squared error (NMSE) of the covariance matrix estimate is defined as:

$$\text{NMSE}(\hat{\mathbf{R}}) = \frac{\|\hat{\mathbf{R}} / \text{Tr}(\hat{\mathbf{R}}) - \mathbf{R} / \text{Tr}(\mathbf{R})\|_F^2}{\|\mathbf{R} / \text{Tr}(\mathbf{R})\|_F^2}, \quad (7)$$

where $\|\cdot\|_F$ denotes the Frobenius norm, $\hat{\mathbf{R}}$ is the estimated value and $\mathbf{R}$ is the true value.

2.2. Toeplitz Hermitian positive definite matrix

The concept of a Toeplitz matrix is first introduced. Consider $T = \{t_{ij}\} \in \mathbb{C}^{n \times n}$, where $t_{ij} = t_{i-j}$ for $i, j = 1, 2, \dots, n$. A matrix T satisfying this condition is referred to as a Toeplitz matrix. Following this, the THPD covariance matrix is defined as a matrix that combines three properties: it is a Toeplitz matrix, a Hermitian matrix, and positive definite. The THPD covariance matrix $\mathbf{C}$ has the following [40]

$$\mathbf{C} = \begin{bmatrix} c_0 & c_{-1} & c_{-2} & \cdots & c_{-n+1} \\ c_1 & c_0 & c_{-1} & \cdots & c_{-n+2} \\ c_2 & c_1 & c_0 & \cdots & c_{-n+3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_{n-1} & c_{n-2} & c_{n-3} & \cdots & c_0 \end{bmatrix}, \quad (8)$$

where $c_i = \overline{c_{-i}}$ to guarantee the Hermitian property, and the matrix is positive definite.

The matrix is derived from the calculation of the reflection coefficient μ_n . $\mathbf{x}$ represent a snapshot containing N elements. Assuming the signal follows an autoregressive (AR) model, the reflection coefficient corresponds to the k th order regression coefficient. This coefficient can be computed using the regularized Burg algorithm, as detailed below [33]

Initialization

$$f_0(k) = b_0(k) = \mathbf{x}(k), \quad k = 0, \dots, N-1, \quad (9)$$

$$P_0 = \frac{1}{N} \sum_{k=0}^{N-1} |\mathbf{x}(k)|^2. \quad (10)$$

for $n = 1$ to $N-1$ do

$$\mu_n = - \frac{\frac{2}{N-n} \sum_{k=n+1}^N f_{n-1}(k) \bar{b}_{n-1}(k-1) + 2 \sum_{k=1}^{n-1} v_k^{(n)} a_k^{(n-1)} a_{n-k}^{(n-1)}}{\frac{1}{N-n} \sum_{k=n+1}^N (|f_{n-1}(k)|^2 + |b_{n-1}(k-1)|^2) + 2 \sum_{k=0}^{n-1} v_k^{(n)} |a_k^{(n-1)}|^2}, \quad (11)$$

with

$$v_k^{(n)} = \psi_1(2\pi)^2(k-n)^2, \quad (12)$$

$$\begin{cases} a_0^{(n)} = 1 \\ a_k^{(n)} = a_k^{(n-1)} + \mu_n \bar{a}_{n-k}^{(n-1)}, \quad k = 1, \dots, n-1, \\ a_n^{(n)} = \mu_n \end{cases} \quad (13)$$

and

$$\begin{cases} f_n(k) = f_{n-1}(k) + \mu_n b_{n-1}(k-1) \\ b_n(k) = b_{n-1}(k-1) + \bar{\mu}_n f_{n-1}(k) \end{cases}, \quad (14)$$

where the reflection coefficient μ_n is calculated in each iteration by combining the previous stage's error and coefficients, with a spectral smoothness constraint to update the autoregressive model parameters, define the smoothness term v_k to control the smoothness of the μ_n and the recursive update of the autoregressive coefficient and defines the recurrence relationship between $f_n(k)$ and $b_n(k)$ in the model.

2.3. VCF on the Grassmann manifold

The collection of all linear subspaces of fixed dimension p in the Euclidean space $\mathbb{R}^n$ forms the Grassmann manifold $Gr(n, p)$, also known as the Grassmannian [41]. Subspaces and Grassmann manifolds play an important role in applications such as data analysis and signal processing.

The union model of linear subspaces is a commonly used signal model. The signal $\mathbf{x}$ in this model is assumed to be a vector in the union of linear subspaces in $\mathbb{R}^N$, defined as

$$\mathbf{x} \in \bigcup_{i=1}^L X_i, \quad X_i = \{\mathbf{x} = X_i \mathbf{a}_i \mid X_i \in \mathbb{R}^{N \times k}, \mathbf{a}_i \in \mathbb{R}^k\}, \quad (15)$$

where the column vectors of the matrix X_i form the basis of the corresponding subspace X_i , and $\dim(X_i) = k < N$, $\mathbf{a}_i$ is a vector in $\mathbb{R}^k$ that generates the $\mathbf{x}$ when multiplied by the matrix X_i .

The Grassmann manifold $Gr(k, N)$ is a topological space [37,38] in which each point corresponds to a k -dimensional linear subspace of $\mathbb{R}^N$. In general, the union of subspaces in (15) is equivalent to a finite set of points in $Gr(k, N)$, namely

$$\zeta(k, N, L) = \{X_1, \dots, X_L\}, \quad (16)$$

where $X_i \in Gr(k, N)$, $i = 1, \dots, L$ represents a point on the Grassmann manifold, $\zeta(k, N, L)$ denotes a discrete set of points on the Grassmann manifold, which is the union of subspaces containing L subspaces.

Volume, as a significant parameter in the Grassmann manifold space, quantifies the degree of separation between two linear subspaces and is closely related to the principal angles between them. Based on the geometric structure of the subspaces, the volume is utilized to derive a VCF, which characterizes the distance between subspaces. The d -dimensional volume of a full rank matrix S is defined as [42]

$$\text{vol}_d(S) = \prod_{i=1}^d \sigma_i, \quad (17)$$

where σ_i , $i = 1, \dots, d$ represent the singular values of the matrix S . Further, for two subspaces whose basis matrices are denoted by $S_1 \in \mathbb{C}^{K \times d_1}$, $S_2 \in \mathbb{C}^{K \times d_2}$, then their volume-dependent functions are defined as

$$\begin{aligned} \text{VCF}(S_1, S_2) &= \frac{\text{vol}_{d_1+d_2}([S_1, S_2])}{\text{vol}_{d_1}(S_1) \text{vol}_{d_2}(S_2)} \\ &= \prod_{i=1}^{\min(d_1, d_2)} \sin \theta_i(\text{span}(S_1), \text{span}(S_2)), \end{aligned} \quad (18)$$

where $\theta_1, \theta_2, \dots, \theta_{\min(d_1, d_2)}$ are the principal angles between the two subspaces spanned by S_1 and S_2 , respectively. The physical meaning is the angle between the d_1 orthonormal vectors in the subspace $\text{span}(S_1)$ and the d_2 orthonormal vectors in the subspace $\text{span}(S_2)$.

3. Proposed BGVCF-STAP algorithm

This section provides a detailed description of the BGVCF-STAP algorithm, with the flowchart shown in Fig. 1. The BGVCF algorithm is primarily divided into three stages: the calculation of the THPD matrix for training samples, the clutter and target screening stage, and the CCM estimation stage.

3.1. Calculation of the THPD matrix

For each range cell, a sliding window of fixed length is applied along the range dimension to form the training sample matrix from the received radar data. The q th training sample is given by $\mathbf{x}(q) = [x_q(1) \ x_q(2) \ \dots \ x_q(MN)]^T$, where $q = 1, 2, \dots, Q$. The training sample matrix is selected using a sliding window of length Q , excluding the test cell and guard cells.

The THPD covariance matrix of each radar snapshot can be estimated by the regularized Burg algorithm. Each THPD covariance matrix represents a convex cone lying on the manifold, enabling distance metric operations.

$$\mathbf{R} = \begin{bmatrix} r_0 & r_1^* & \dots & r_{MN-1}^* \\ r_1 & r_0 & \dots & r_{MN-2}^* \\ \vdots & \vdots & \ddots & \vdots \\ r_{MN-2} & r_{MN-3} & \dots & r_1^* \\ r_{MN-1} & r_{MN-2} & \dots & r_0 \end{bmatrix}, \quad (19)$$

where each r_i can be calculated using the reflection coefficient μ_i , as given by

$$\begin{cases} r_0 = P_0, \quad r_1 = -P_0 \mu_1 \\ r_i = -P_{i-1} \mu_i \alpha_{i-1}^T \mathbf{J}_{i-1} \mathbf{R}_{i-1}^{-1} \alpha_{i-1}, \quad 2 \leq i \leq MN-1 \end{cases}, \quad (20)$$

P_0 and μ_i are obtained through the regularized Burg algorithm (11)–(16), this illustrates how the recursive process updates the current r_i by combining the previous calculation results [33].

$$\alpha_{i-1} = \begin{bmatrix} r_1 \\ \vdots \\ r_{i-1} \end{bmatrix}, \quad \mathbf{J}_0 = \mathbf{I}, \quad \mathbf{J}_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad (21)$$

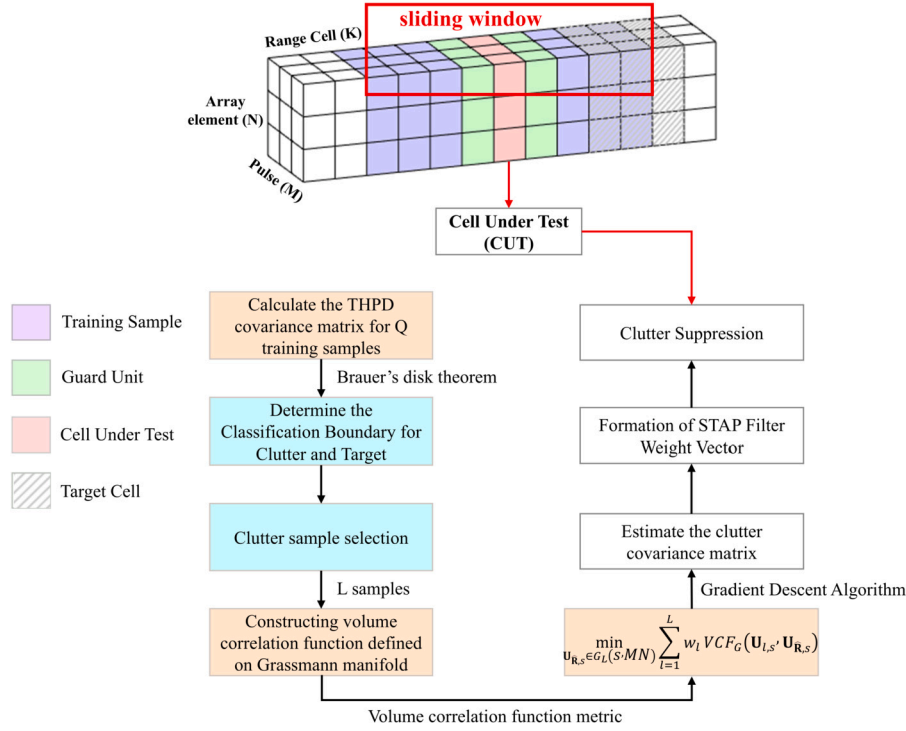


Fig. 1. Algorithm flowchart.

$$\mathbf{J}_{i-1} = \begin{bmatrix} 0 & 0 & \dots & 1 \\ 0 & \dots & 1 & 0 \\ \vdots & \dots & \dots & \vdots \\ 1 & \dots & 0 & 0 \end{bmatrix}, \quad (22)$$

$$\mathbf{R}_{i-1} = \mathbf{R}(2 : i, 1 : k-1), \quad P_{i-1} = P_0 \prod_{j=1}^{i-1} (1 - |\mu_j|^2). \quad (23)$$

Through the above definition, the THPD covariance matrix of Q training samples is calculated.

3.2. Distinguish clutter-plus-noise and targets

Theorem 1. Brauer Disk Theorem [35]: Suppose $R = [r_{ij}]$ is an n -th order complex Toeplitz matrix, with $r_{ii} = r_0$, and $n \geq 2$. Let $\sigma(R)$ denote the eigenvalues of R . Then,

$$\sigma(R) \subseteq Y(R) = \bigcup_{i=1}^{\left\lfloor \frac{n}{2} \right\rfloor} Y_i(R), \quad (24)$$

$$Y_i(R) = \{z \in \mathbb{C} : |z - r_0| \leq D\}, \quad (25)$$

$$D = \max_{i,j \in N, i \neq j} \sqrt{a_i(R) a_j(R)}, \quad (26)$$

$$a_i(R) = \sum_{j \neq i} |r_{ij}|, \quad (27)$$

$$\left\lfloor \frac{n}{2} \right\rfloor = \begin{cases} \frac{n}{2}, & n = 2k, k \in \mathbb{Z} \\ \frac{n+1}{2}, & n = 2k+1, k \in \mathbb{Z} \end{cases}, \quad (28)$$

where $Y(R)$ is the union of the Brauer circle sets of the symmetric matrix R .

The BD theorem is a mathematical tool [36] used to determine the inclusion region of eigenvalues for a square matrix. It takes into account both the sum of the elements in the rows and the sum of the elements in the columns of the matrix. This enables it to define the location of the eigenvalues more precisely, providing tighter bounds. Based on BD theorem, the clutter and target signal THPD covariance matrix are classified. The basic principle is as follows:

- (1) Eigenvalue analysis: each THPD covariance matrix is decomposed by eigenvalue to obtain its eigenvalue. The center of the largest BD, the contained region of the largest eigenvalues, P_0 represents the snapshot power, as shown in (10).
- (2) Determination of the distribution range of eigenvalues: according to the distribution range of eigenvalues, the minimum BD radius ρ of the maximum eigenvalue of each THPD covariance matrix is calculated. This radius describes the distribution range of eigenvalues in the complex plane, reflects the degree of disturbance of the matrix, and represents the minimum limit of clutter-plus-noise.
- (3) Determination of classification limits: using the BD centers of all THPD covariance matrices, the Brauer cluster boundary limit T_B for clutter-plus-noise can be established [43]

$$T_B = \frac{\frac{1}{K} \sum_{k=1}^K \hat{\mathbf{R}}_k}{\left(\prod_{k=1}^K \hat{\mathbf{R}}_k \right)^{\frac{1}{K}}} \rho, \quad (29)$$

where $\frac{1}{K} \sum_{k=1}^K \hat{\mathbf{R}}_k$ and $\left(\prod_{k=1}^K \hat{\mathbf{R}}_k \right)^{\frac{1}{K}}$ represent the arithmetic mean and geometric mean of the BD centers, respectively. In general, the eigenvalue distribution range of a clutter matrix is relatively small, resulting in a smaller minimum BD radius. However, for matrices containing target signals, the situation is the opposite.

- (4) Classification and screening: The minimum BD radius of all THPD covariance matrices is compared with the classification threshold T_B , and matrices with a radius smaller than T_B are considered as clutter matrices. Specifically, the THPD matrices corresponding to eigenvalues within the Brauer cluster boundary are regarded as clutter-plus-noise covariance matrices, while those corresponding to eigenvalues outside the Brauer cluster boundary are considered as target matrices.

3.3. Estimation of CCM based on GVCF

In general, the CCM estimation for clutter suppression can be viewed as a function of the matrix composed of the received data in the training data set

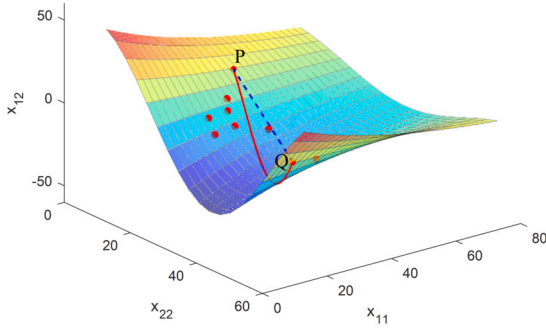


Fig. 2. Schematic diagram of distance measurement on manifolds.

$$\hat{\mathbf{R}} = \phi(\hat{\mathbf{R}}_1, \hat{\mathbf{R}}_q, \dots, \hat{\mathbf{R}}_Q) = \phi'(x_1, x_q, \dots, x_Q), \quad (30)$$

where $\hat{\mathbf{R}}_q$ represent the THPD covariance matrix of the q th training sample, and $\hat{\mathbf{R}}$ be the estimated CCM. $x_q = x(q)$ denotes the training data without the target. The CCM can be estimated using the following optimization model

$$\min_{\hat{\mathbf{R}} \in \mathbb{R}} \sum_{q=1}^Q w_q d_E^2(\hat{\mathbf{R}}_q, \hat{\mathbf{R}}), \quad (31)$$

where $\mathbb{R}$ represents the linear space spanned by the observation matrices, $w_q > 0$ are the weights that satisfy $\sum_{q=1}^Q w_q = 1$, indicating the proportion of each training unit in the estimation. $d_E^2(\hat{\mathbf{R}}_q, \hat{\mathbf{R}})$ represents the Euclidean distance between the two spaces.

However, the Euclidean distance only focuses on the signal energy and does not fully exploit the structural information in the data. Specifically, the Euclidean distance $d_E(\hat{\mathbf{R}}_q, \hat{\mathbf{R}})$ calculates the discrepancy between two matrices using the Frobenius norm, which is defined as

$$d_E^2(\hat{\mathbf{R}}_q, \hat{\mathbf{R}}) = \|\hat{\mathbf{R}}_q - \hat{\mathbf{R}}\|_F^2 = \text{Tr}((\hat{\mathbf{R}}_q - \hat{\mathbf{R}})(\hat{\mathbf{R}}_q - \hat{\mathbf{R}})^H), \quad (32)$$

where $\|\cdot\|_F$ denotes the Frobenius norm, $\text{Tr}(\cdot)$ represents the trace of the matrix, this measure overlooks the internal geometric and structural properties of the CCM.

Fig. 2 illustrates distance measurement on a manifold, highlighting the limitations of the Euclidean metric in high-dimensional structures due to its neglect of geometric information [44]. The red circles represent positive definite covariance matrices, with the axes x_{11} , x_{22} , and x_{12} visualizing the matrix space. In the figure, the Euclidean distance (blue dashed line) fails to accurately reflect the true distance between points P and Q on the manifold, while the red geodesic path represents the actual intrinsic distance [45]. The limitation of the Euclidean distance lies in its inability to preserve the positive definiteness of matrices. For instance, when a positive definite matrix is scaled by a negative scalar, the result is no longer positive definite. Therefore, using the Euclidean distance to measure THPD covariance matrices is inappropriate, whereas the structure of the Grassmann manifold naturally preserves this characteristic.

The Grassmann manifold $G(k, n)$ is the collection of all k -dimensional subspaces within an n -dimensional space. In other words, the Grassmann manifold is the set of all $n \times k$ matrices, where each $n \times k$ matrix represents a k -dimensional subspace, and these matrices are orthonormal. For Q THPD matrices of size $MN \times MN$, eigenvalue decomposition or singular value decomposition is performed. Since $\hat{\mathbf{R}}_q$ is Hermitian positive definite, it can be expressed as

$$\hat{\mathbf{R}}_q = \mathbf{U}_q \mathbf{\Lambda}_q \mathbf{U}_q^H, \quad (33)$$

where $\mathbf{U}_q$ is the eigenvector matrix of the q -th THPD matrix of size $MN \times MN$, and $\mathbf{\Lambda}_q$ is the diagonal matrix of eigenvalues of the q -th THPD matrix. Each diagonal element corresponds to an eigenvalue. From each $\mathbf{U}_q$, the first s eigenvectors (corresponding to the largest s eigenvalues) are selected to form an $MN \times s$ matrix $\mathbf{U}_{q,s}$. The column

space of this matrix is an s -dimensional subspace. Each THPD matrix $\hat{\mathbf{R}}_q$ is mapped to an s -dimensional subspace, and the collection of these subspaces forms a point on the Grassmann manifold $G_Q(s, MN)$.

Based on the structure of the Grassmann manifold $G_Q(s, MN)$, the volume cross-correlation function on the Grassmann manifold can be used to measure the similarity between $\hat{\mathbf{R}}_i$ and $\hat{\mathbf{R}}_j$

$$d_G(\hat{\mathbf{R}}_i, \hat{\mathbf{R}}_j) = \text{VCF}_G(\mathbf{U}_{i,s}, \mathbf{U}_{j,s}) = \frac{\text{vol}_{2s}([\mathbf{U}_{i,s}, \mathbf{U}_{j,s}])}{\text{vol}_s(\mathbf{U}_{i,s})\text{vol}_s(\mathbf{U}_{j,s})}, \quad (34)$$

where

$$\text{vol}_s(\mathbf{U}_{q,s}) = \prod_{l=1}^s \delta_l, \quad (35)$$

and $\delta_1, \delta_2, \dots, \delta_s \geq 0$ are the singular values of $\mathbf{U}_{q,s}$. Since $\mathbf{U}_{q,s}$ is an orthonormal basis, by substituting $\text{vol}_s(\mathbf{U}_{q,s}) = \prod_{l=1}^s \delta_l = 1$, the distance between two subspaces can be rewritten as

$$d_G(\hat{\mathbf{R}}_i, \hat{\mathbf{R}}_j) = \text{VCF}_G(\mathbf{U}_{i,s}, \mathbf{U}_{j,s}) = \text{vol}_s([\mathbf{U}_{1,s}, \mathbf{U}_{2,s}]). \quad (36)$$

Finally, the CCM can be estimated using the following optimization model

$$\min_{\mathbf{U}_{\mathbf{R},s} \in G_Q(s, MN)} \sum_{q=1}^Q w_q \text{VCF}_G(\mathbf{U}_{q,s}, \mathbf{U}_{\mathbf{R},s}), \quad (37)$$

where $\hat{\mathbf{R}} = \mathbf{U}_{\mathbf{R}} \mathbf{\Lambda}_{\mathbf{R}} \mathbf{U}_{\mathbf{R}}^H$ and $\mathbf{U}_{\mathbf{R},s}$ is the s -dimensional subspace of $\mathbf{U}_{\mathbf{R}}$. Although the VCF only depends on the eigenvector, to obtain a complete covariance matrix, it is still necessary to ensure that the eigenvalue matrices $\mathbf{\Lambda}_{\mathbf{R},s}$ and $\mathbf{\Lambda}_{\mathbf{R}}$ are updated.

Algorithm 1 BGVCf-STAP Algorithm.

Input: STAP data $\mathbf{x}$, number of samples q , gradient descent step size η

Output: STAP suppression results $\mathbf{y}$

- 1: Calculate THPD matrix $\mathbf{R}_q$
 - 2: Establish Brauer cluster boundary limit T_B
 - 3: Obtain clutter training samples $\hat{\mathbf{R}}_q$
 - 4: Eigenvalue decomposition on $\hat{\mathbf{R}}_q = \mathbf{U}_q \mathbf{\Lambda}_q \mathbf{U}_q^H$
 - 5: Select eigenvectors from $\mathbf{U}_q$ to form $\mathbf{U}_{q,s}$, $\mathbf{\Lambda}_{q,s}$ and initialize $\hat{\mathbf{R}}$
 - 6: Update $\mathbf{U}_{\mathbf{R},s}$ using gradient descent $\mathbf{U}_{\mathbf{R},k} \leftarrow \mathbf{U}_{\mathbf{R},k} - \eta \frac{\partial \text{VCF}}{\partial \mathbf{U}_{\mathbf{R},k}}$
 - 7: After the update, recalculate the covariance matrix $\hat{\mathbf{R}} = \mathbf{U}_{\mathbf{R},s} \mathbf{\Lambda}_{\mathbf{R},s} \mathbf{U}_{\mathbf{R},s}^H$
 - 8: Space-time filter $\mathbf{w} = \frac{\hat{\mathbf{R}}^{-1} \mathbf{s}}{\mathbf{s}^H \hat{\mathbf{R}}^{-1} \mathbf{s}}$
 - 9: $\mathbf{y} = \mathbf{w}^H \mathbf{x}$
-

3.4. Computational complexity

The computational complexity of the traditional STAP method is $\mathcal{O}(Q(MN)^3)$, which is mainly dominated by the inversion of the covariance matrix and sample estimation. In addition, the other STAP comparison methods mentioned in this paper have the same computational complexity as traditional STAP, $\mathcal{O}(Q(MN)^3)$, or require a slightly increased computational cost. Then, the new method proposed in this paper can be divided into two parts in terms of computational complexity. One part is the calculation and eigendecomposition of the THPD matrix, which has a computational cost of $\mathcal{O}(Q(MN)^3)$; the other part is the VCF calculation and optimization, $\mathcal{O}(TQMNs^2)$, so the total computational cost is $\mathcal{O}(Q(MN)^3 + TQMNs^2)$, where Q is the number of training samples, $T(T \ll MN)$ is the number of optimization iterations, $s(s \ll MN)$ is the subspace dimension.

4. Experimental result and analysis

In this section, the proposed BGVCf-STAP method is evaluated and compared with several conventional approaches, including loaded sample matrix inversion (LSMI), GIP, RMT, MAP and the GVCf-STAP

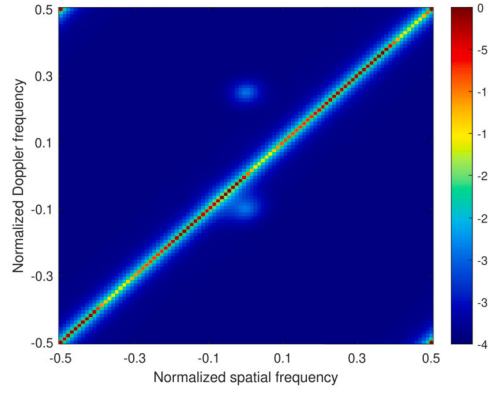


Fig. 3. Spatial-temporal Capon spectrum.

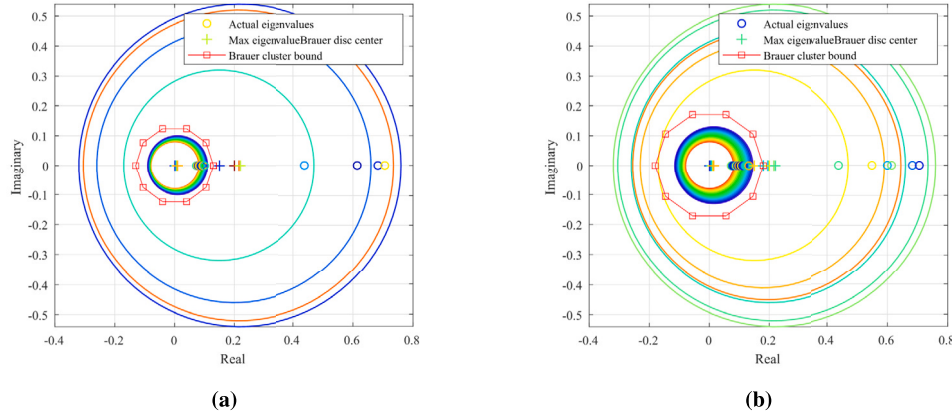


Fig. 4. Brauer bound for clutter-plus-noise and targets under different numbers of training samples. (a) number of samples = 25. (b) number of samples = 50.

Table 1
Simulation Parameters.

Parameter	Value
Carrier frequency	1.5 GHz
PRF	3000 Hz
Platform velocity	150 m/s
Platform height	10 km
Number of pulses	12
Number of antenna elements	10
Bandwidth	10 MHz
Number of azimuthal clutter patches	201
Clutter-to-noise ratio	50 dB

method without applying the BD theorem. Both simulated data and Mountain-Top measured data are utilized to demonstrate the effectiveness of proposed algorithm.

4.1. Simulated data

This section presents the evaluation of the proposed algorithm's performance using simulated data. The airborne radar system is modeled as a side-looking ULA, with array elements spaced at half the wavelength. The key simulation parameters are summarized in Table 1. A heterogeneous clutter environment is simulated, and to account for the effect of internal clutter motion, the range-spread target 1 is introduced in range cells 30 to 32, with normalized Doppler and spatial frequencies of 0.25 and 0, respectively. Similarly, the range-spread target 2 is introduced into range cells 39–41, with normalized Doppler and spatial frequencies of -0.1 and 0, respectively. The Spatial-temporal Capon spectrum is shown in Fig. 3.

The effectiveness of the proposed techniques was assessed through two separate tests, each using 25 and 50 snapshots for analysis, respec-

tively. The selected snapshots were transformed into THPD covariance matrices, and a clutter-plus-noise bound was determined using the BD theorem. As illustrated in Fig. 4, some points fall outside the established clutter-plus-noise bound, indicating their potential as target candidates. In Fig. 4 (a), the range cells from 15 to 39 were extracted, and four target points (30–32, 39) were detected outside the boundary. In Fig. 4 (b), the range cells from 15 to 64 were extracted, and all six target points were detected.

Fig. 5 illustrates the spatial slice and Doppler slice beampatterns, which are extracted from the space-time plane beampatterns of six algorithms at the target Doppler and spatial frequencies. These patterns are used to compare the performance details of the algorithms. Fig. 5 (a) and Fig. 5 (b) show the spatial slice beampatterns at the Doppler frequencies of target 1 and target 2, respectively. Traditional methods such as LSMI and GIP exhibit higher sidelobes, whereas the proposed BGVCf method achieves lower sidelobes compared to the GVCf algorithm. Additionally, the clutter notches of the BGVCf method are deeper than those of other methods. In Fig. 5 (c), which depicts the Doppler slice beampattern, the proposed method also demonstrates lower sidelobes and deeper clutter notches than other algorithms. These results clearly indicate that the proposed method significantly outperforms others in clutter suppression, ensuring better consistency between the target location and the preset position.

As shown in Fig. 6, the Capon spectra calculated using different CCM estimation methods exhibit significant performance differences. From (a) to (f), the spectral performance progressively improves. Among them, the LSMI method has poor clutter estimation accuracy, with more background noise and false peaks; the GIP method shows some improvement over LSMI but still has significant deviations; the RMT method further enhances performance, with a clearer clutter ridge; the MAP method results in progressively converging clutter features; the GVCf-

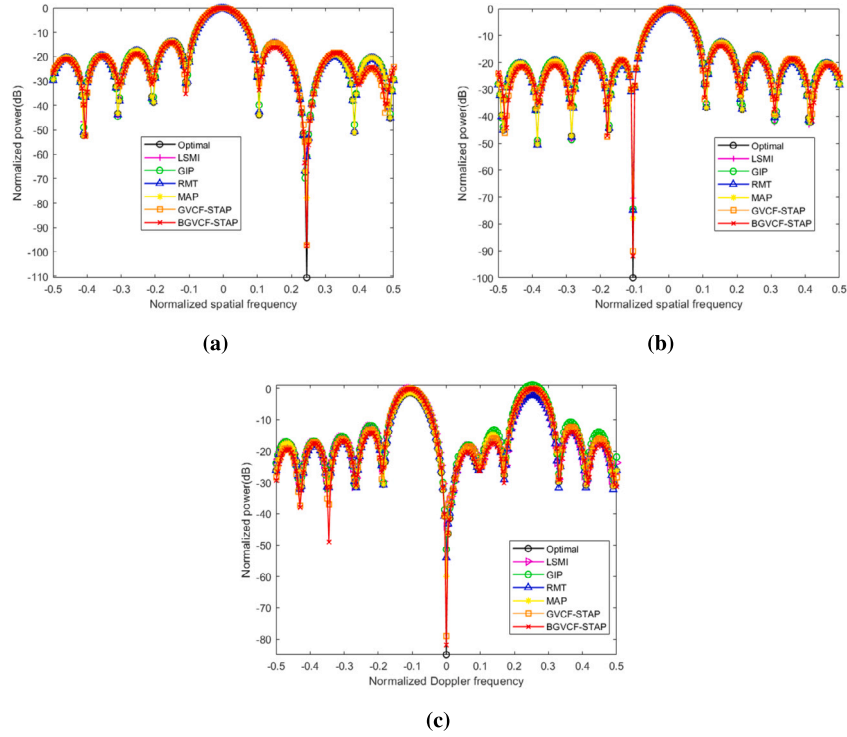


Fig. 5. Spatial and Doppler slice beampattern. (a) spatial slice beampattern of target1. (b) spatial slice beampattern of target2. (c) Doppler slice beampattern.

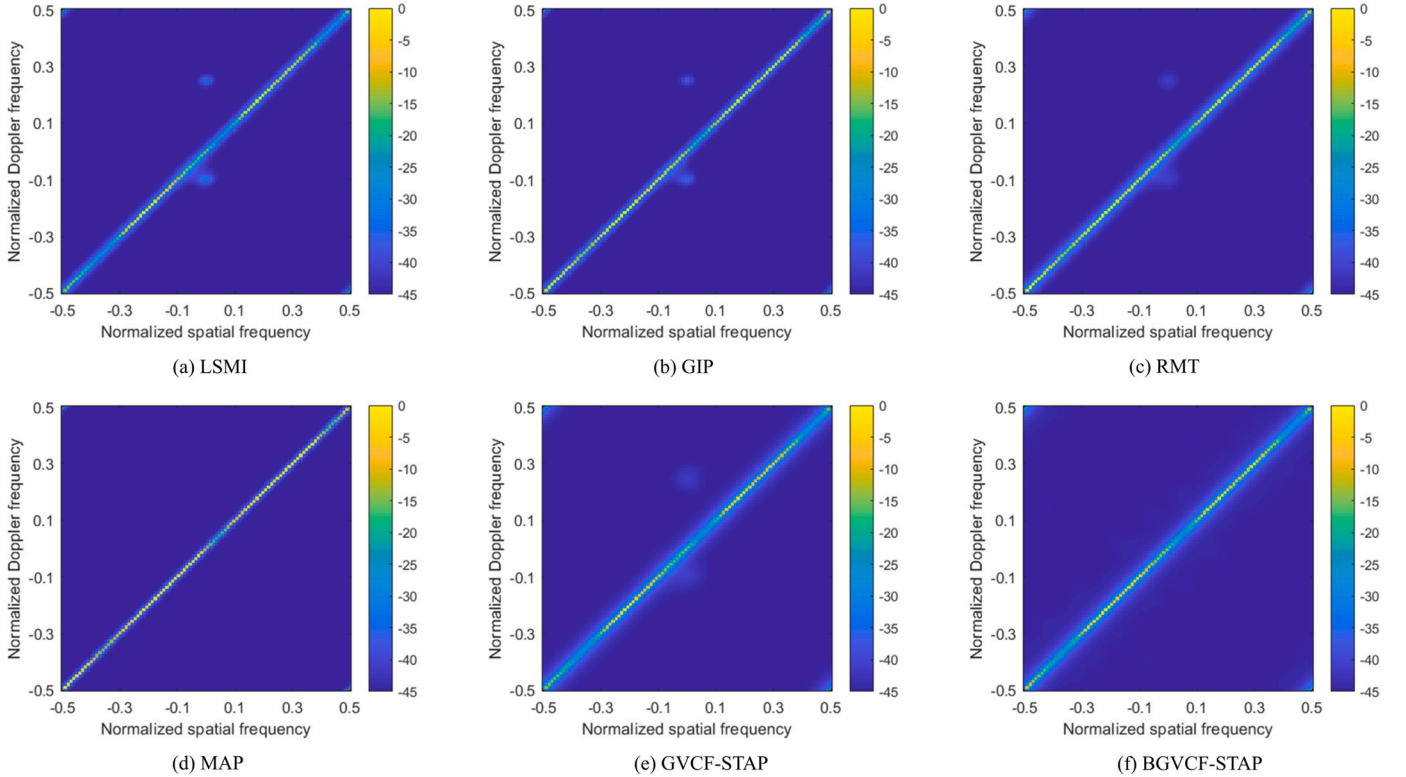


Fig. 6. Capon spectra calculated with different CCM estimations.

STAP method achieves relatively ideal clutter suppression, and the spectrum approaches the ideal state; the BGVCVF-STAP method performs the best, with significant values strictly located along the clutter ridge, and the clutter estimation reaches its optimal state. This trend indicates that as the CCM estimation method improves, the quality of the Capon spec-

trum is significantly enhanced. In order to better analyze the accuracy of clutter covariance matrix estimation, Fig. 7 shows the NMSE of CCM estimation under different sample numbers.

The improvement factor (IF) curve [46] of different algorithms are shown in Fig. 8. The IF curve of the proposed BGVCVF-STAP algorithm

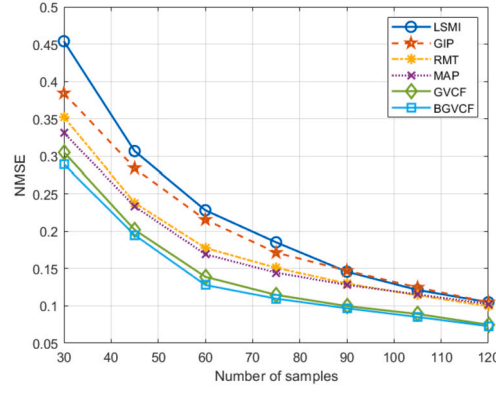


Fig. 7. Estimation accuracy of the clutter covariance matrix.

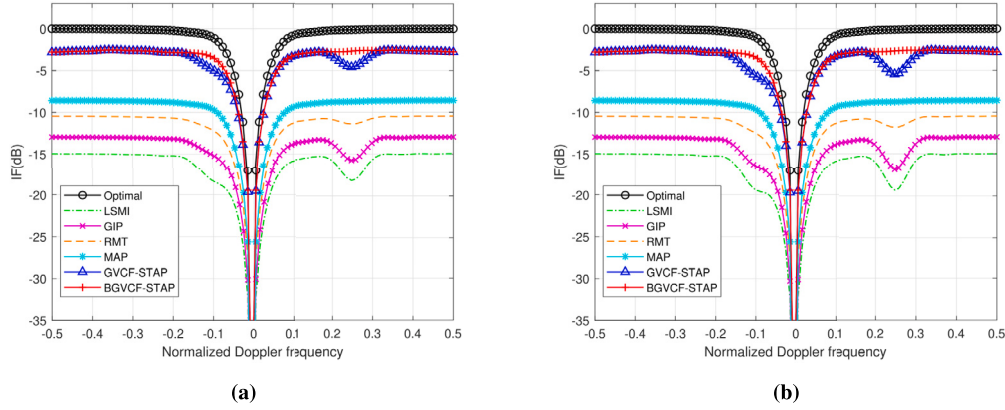


Fig. 8. Comparison of IF under different SNR. (a) SNR = -5 dB. (b) SNR = 0 dB.

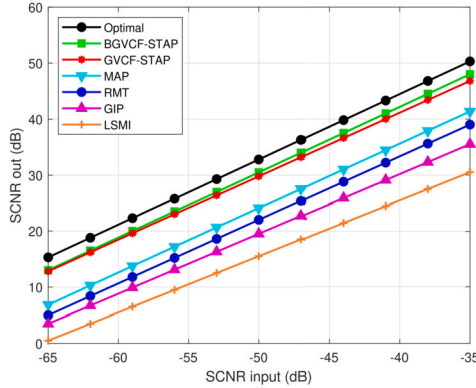


Fig. 9. Clutter suppression performance under different input SCNR.

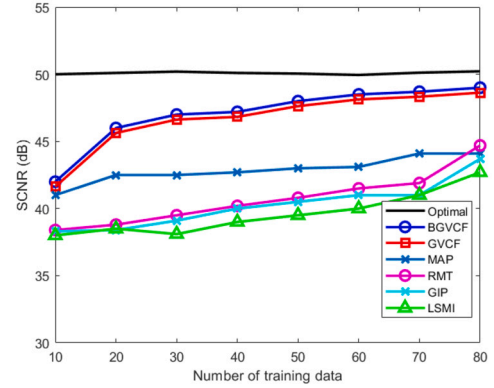


Fig. 10. Output SCNR against the number of training data.

closely approximates the optimal curve, outperforming the MAP method by more than 5 dB and exceeding the LSMI method by over 12 dB. This is attributed to its effective utilization of the internal geometric structure of the data, without requiring any prior knowledge of the clutter distribution. GVCVF-STAP algorithm suffers from performance degradation due to the presence of targets in the training sample.

Fig. 9 shows the output signal-to-clutter-plus-noise ratio (SCNR) [47] of different algorithms under varying input SCNR conditions. BGVCVF-STAP performs the best, closely approximating the optimal case. Due to the presence of targets in the training samples, GVCVF-STAP performs slightly worse than BGVCVF-STAP as SCNR increases. MAP ranks third, as it effectively models the actual clutter distribution. Both RMT and GIP methods are able to suppress interference from range-spread targets, but they perform worse than MAP, while the LSMI method shows the poorest performance. Overall, BGVCVF-STAP and GVCVF-STAP are partic-

ularly effective under low input SCNR conditions, making them suitable for scenarios that require strong clutter suppression. To further evaluate the potential of the proposed algorithm, Fig. 10 presents the relationship between the output signal clutter-to-noise power ratio (SCNR) and the number of training samples for different methods. As seen in Fig. 10, the superiority of the proposed method is evident from the start. Subsequently, as the number of training samples increases, the performance of all methods tends to converge.

4.2. Measured data

We then apply the BGVCVF-STAP algorithm to the publicly available Mountain-Top dataset [48]. This dataset includes a receive antenna array with 14 elements, where the echoes from 16 coherent pulses are grouped to form a CPI data set. The system used a 500 kHz linear

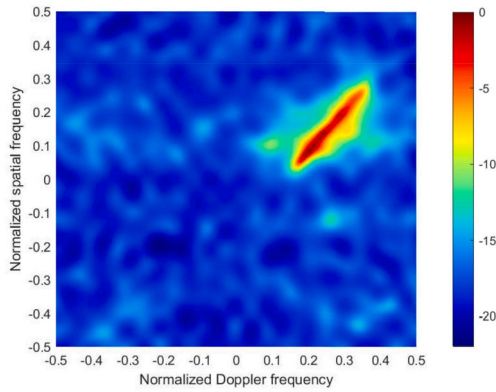


Fig. 11. Estimated clutter spectrum using all 403 range snapshots.

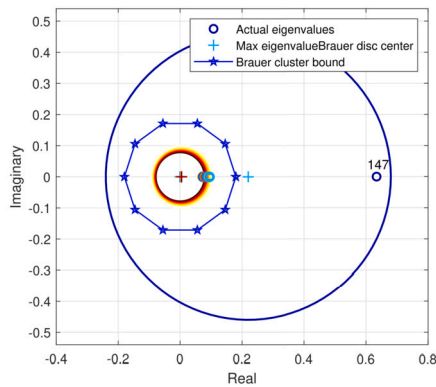


Fig. 12. Brauer bound for clutter-plus-noise and target in Mountain-Top data.

frequency-modulated pulse for transmission, with a pulse width of 100 μ s and a pulse repetition frequency (PRF) of 625 Hz. For our experiment, we utilize the data file t38pre01v1 CPI6, which contains 403 range snapshots. In this dataset, the target is positioned in the 147th range cell, with a normalized Doppler frequency of 0.25 and a normalized spatial angle of -15° . The clutter power spectrum, estimated from all 403 range cells, is shown in Fig. 11.

We used the BD theorem to establish bounds for the measured data in the 110th to 180th range cells, resulting in the outcomes shown in Fig. 12. As observed, a small circle appears outside the Brauer bound for clutter-plus-noise and target, indicating that a target is detected at the 147th range cell. This finding is consistent with the known information: the target in the Mountain-Top data is located at the 147th range cell with a normalized Doppler frequency of 0.25. In this paper, we employ a sliding window approach. For each CUT, we select 40 surrounding cells as training samples. Additionally, two guard cells are placed on both sides of the CUT. Fig. 13 illustrates the STAP output power across the 110th to the 180th range cells. As depicted in the figure, the highest output power is observed at the 147th range cell, where the target is located. In contrast, the output power in the neighboring range cells, processed by the proposed BGVCf-STAP algorithm, remains low, thereby enhancing target detection. Table 2 shows the average power of clutter-plus-noise after clutter suppression processing by each method, which more intuitively reflects the output SCNR performance of each method.

5. Conclusion

This paper presents a novel STAP clutter suppression method based on the BD theorem and GVCf. The proposed method uses a THPD covariance matrix to smooth the data, reduce variance, and enhance the consistency between the covariance matrix and the actual signal

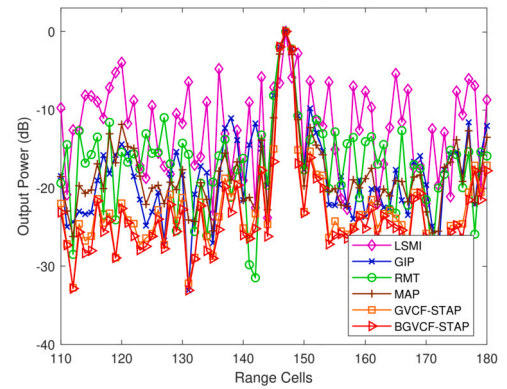


Fig. 13. Output power of different methods.

Table 2

Average Power of Clutter-Plus-Noise for Different Methods.

Method	Average Power
LSMI	-12.9097 dB
GIP	-19.4430 dB
RMT	-18.0040 dB
MAP	-19.3090 dB
GVCf-STAP	-23.8631 dB
BGVCf-STAP	-24.9755 dB

model, thereby improving estimation accuracy and algorithm performance. By applying the Brauer clustering boundary derived from the BD theorem, it effectively distinguishes range-spread targets from clutter. The core of this method lies in transforming the clutter suppression problem into a target-clutter discrimination problem on the Grassmann manifold. By leveraging the VCF and gradient descent optimization, the method utilizes the geometric structure of the manifold space to estimate the CCM. Simulations and measured data validate the applicability of this approach in complex and heterogeneous environments. The results demonstrate that the proposed method outperforms existing techniques in clutter suppression performance. In summary, this study introduces a new VCF clutter suppression method based on manifold geometry. The algorithm enhances radar detection capabilities in challenging scenarios and lays the foundation for further exploration of STAP methods on the Grassmann manifold. In future research, we will consider combining other structured matrix algorithms with the geometric properties of the Grassmann manifold to improve the robustness of covariance estimation in heterogeneous environments.

CRedit authorship contribution statement

Jia-Mian Li: Data curation, Methodology, Software, Validation, Writing – original draft. **Jian-Yi Chen:** Methodology, Validation, Writing – review & editing. **Bing-Zhao Li:** Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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